KSISU Skillathon 2026

A Reproducible Method for Identifying Where District Data Is Missing - and Which District to Onboard First

Manufacturing pilot • NCO 72 / 74 / 81 • 2022-23 reference year

The proposal does not claim an official Karnataka Skill Index formula. It demonstrates a transparent public-data method that separates observed inputs from proxies, propagates uncertainty, and turns uncertainty itself into an onboarding decision.

Core output: a district priority list for where KSISU should acquire better district trade-demand and ITI-supply data first.

1. The problem is not the dashboard

Karnataka has multiple labour-market data streams, but they do not share the same grain, identifiers, time basis, or completeness. The challenge therefore is not to merge two clean tables; it is to build a defensible evidence chain from heterogeneous public sources to a policy action.

This pilot narrows the analytical question to a tractable use case: compare ITI trade supply with manufacturing occupation openings for NCO groups 72, 74 and 81. When district-actual supply is missing, the system does not invent it; it estimates a proxy, exposes the uncertainty, and asks where better data would have the largest decision value.

Layer	What is observed / estimated
Source registry	file_id, SHA-256, source, period, grain, version
District key	LGD code + alias + fuzzy match + human review
Demand	State manufacturing base + district proxy allocation + NCO mix
Supply	District trade seats if available; otherwise transparent proxy
Inference	5,000 shared-draw Monte Carlo alignment ratio
Decision	Proxy-only -> NEEDS DATA; actual supply -> provisional gap/surplus/data-needed
Onboarding	VOI = interval width × annual openings

2. Harmonisation is a scored method, not a string replace

District matching is a cascade: exact normalised name → known alias → fuzzy match at 0.90 → human review. Every match records the method and confidence. Unmatched rows are never dropped.

Step	Example	Output
Exact	Bengaluru Urban	LGD code + method=exact
Alias	Gulbarga → Kalaburagi	LGD code + method=alias
Fuzzy	Chikkaballpura → Chikkaballapur	LGD code + method=fuzzy
Review	Unknown district label	unmatched.csv + human review queue

For occupations, the pilot restricts the analysis to NCO groups 72, 74 and 81. A versioned course/trade → NCO crosswalk is used for ITI supply. When PLFS unit-level occupation-by-industry data is available, the analyst prior can be replaced without changing the downstream model.

3. Demand model: one base, one allocation rule

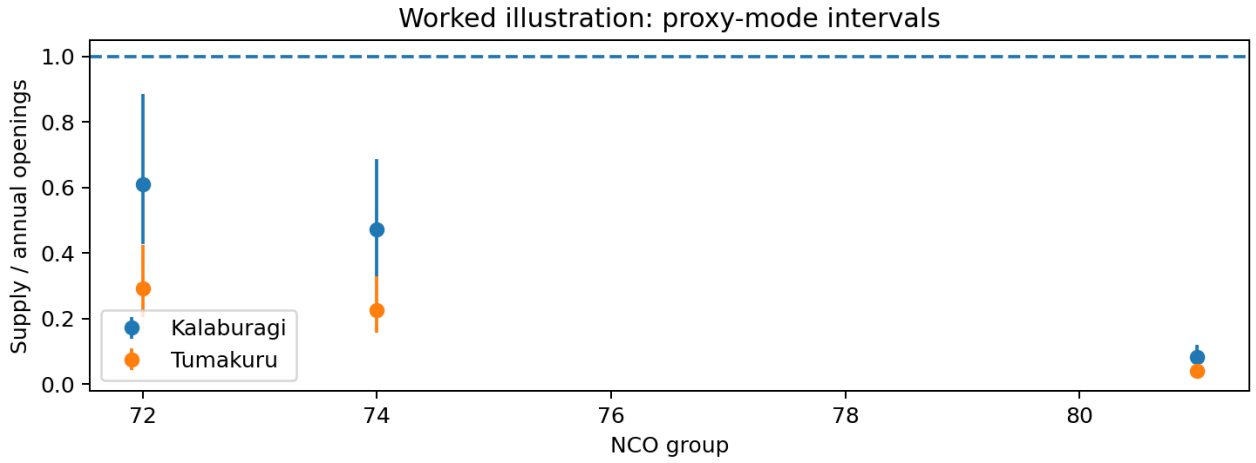
State manufacturing base: Karnataka selected-manufacturing persons engaged in ASI 2022-23 = 1,166,555.

District proxy: allocate that state manufacturing base using the district share of Udyam registered-employment. This is explicitly a proxy because Udyam employment is not a manufacturing-only labour census.

Occupation mix: restrict to NCO 72/74/81 and normalise the versioned prior. Replace with observed PLFS unit-level occupation-by-industry shares when available.

Annual openings: opening rate is triangular 5%-8.5%-12%.

Quantity	Formula
District manufacturing employment proxy	M_d = M_state × UdyamShare_d
Occupation stock	E_d,o = M_d × m_o
Annual openings	O_d,o = E_d,o × r_open
Alignment ratio	R_d,o = S_d,o / O_d,o



4. Supply model: seats are not graduates

The supply equation keeps flows separate: **Seats × Fill × Completion × Sector entry**. A seat is never counted as a graduate, and completions are never added to seats.

Mode	Seat source	Evidence
Proxy mode	NCVT MIS Karnataka state seats + DGT division structure + district Udyam allocation	transparent but not district-actual
District-actual mode	NCVT MIS / DGT / Karnataka district-trade extract loaded by user	replaces proxy seats for those districts

The public NCVT MIS ITI Count page reports Karnataka with 1,467 ITIs and 216,772 seats available in 2022. NCVT MIS also exposes district/trade/seat fields in downloadable ITI extracts, which is the preferred input for the live demonstration.

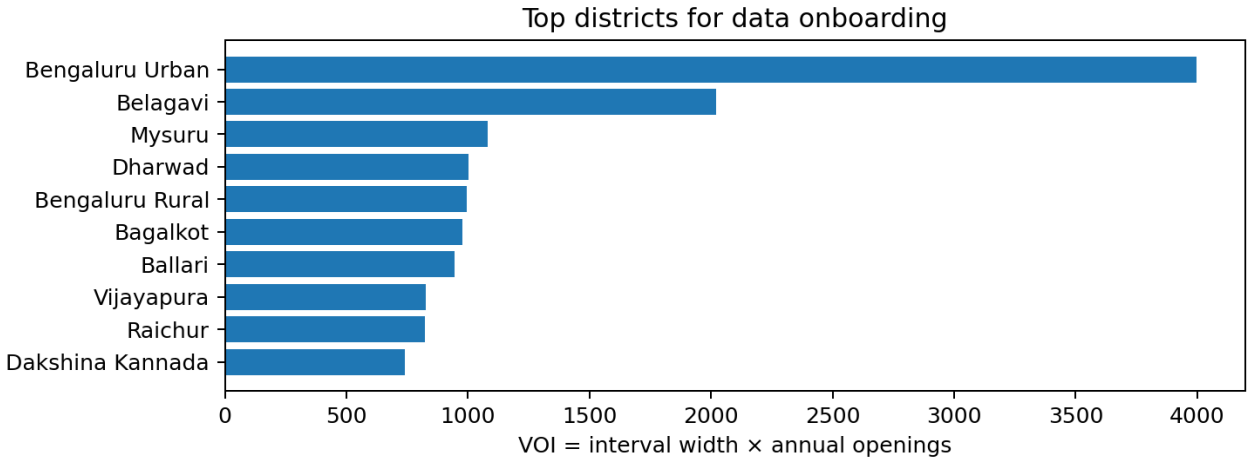
Proxy-only results are not treated as empirical district gaps. They are used to rank the next data collection action.

5. Policy rule: do not overclaim the proxy run

The current bundled run contains 93 district×occupation rows. The mathematical interval identifies 86 cells with p90<1, but because district supply is still proxy-based, the policy-facing status remains NEEDS DATA. This is intentional.

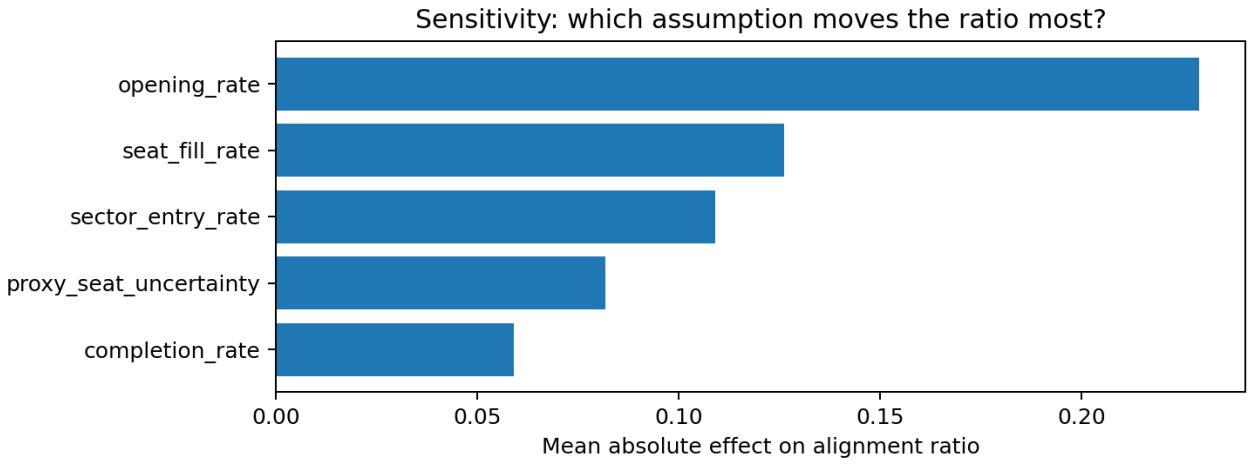
Once a district-actual ITI trade file is loaded, the same model can produce PROVISIONAL GAP / PROVISIONAL SURPLUS / NEEDS DATA for those districts. That creates a visible bridge from public-data inference to real KSISU onboarding.

Rank	District	SCI	VOI	Confidence
1	Bengaluru Urban	0.191	3,998	65%
2	Belagavi	0.465	2,021	56%
3	Mysuru	0.264	1,080	62%
4	Dharwad	0.465	1,002	56%
5	Bengaluru Rural	0.191	996	65%
6	Bagalkot	0.465	976	56%
7	Ballari	0.400	946	58%
8	Vijayapura	0.465	827	56%
9	Raichur	0.400	822	58%
10	Dakshina Kannada	0.264	740	62%



6. Sensitivity: know what assumption matters

The model changes one assumption at a time from low to high while holding other central values fixed. The current pilot indicates the opening-rate range is the strongest driver, followed by seat fill and sector-entry. This tells KSISU which additional evidence is most valuable to improve the decision.



Parameter	Mean absolute ratio effect
opening_rate	0.229
seat_fill_rate	0.126
sector_entry_rate	0.109
proxy_seat_uncertainty	0.082
completion_rate	0.059

7. Validation and reproducibility

Test	Current result	Interpretation
District corruption recovery	96.8%	Recovered 30/31 districts
Duplicates removed	2	5% duplicate injection test
Temporal sanity	0.25 pp median	temporal consistency check, not an independent validation benchmark
Randomness	5,000 shared draws	same draw index across districts
Lineage	SHA-256 + source file IDs	reproducible raw-to-output chain

Important: the temporal Udyam-share check is a consistency test, not an independent ground-truth validation. An independent district total can be supplied through the GUI validation template.

8. What the judges see: live user-input GUI

The interface is designed for a three-minute demonstration. The judge can see the assumptions, swap in a real district-trade file, rerun the identical analysis, and inspect the resulting evidence state.

Action	What you say	What changes
Run bundled research inputs	"This is the reproducible public-data baseline."	Proxy mode; policy stays NEEDS DATA
Show District Priority	"The model tells KSISU where better data matters most."	VOI ranking
Load NCVT MIS district-trade export	"Now I replace one proxy with actual district supply."	Supply basis becomes district-actual
Rerun	"The uncertainty should shrink because one unknown has been measured."	P10/P90 width changes
Open Sensitivity	"We can see which assumption still drives uncertainty."	parameter ranking
Open Validation	"Nothing disappears: mismatches and duplicates stay visible."	schema / duplicate test

User input file shape: **district, nco_group_2digit, seats** or **district, trade, seats**. The template is included in `input_templates/`. Use only downloaded/public values in the live demonstration.

9. Sources, safeguards and next build

Primary public sources used or supported by the pipeline:

LGD Download Directory
https://lgdirectory.gov.in/demo/downloadDirectory.do

Udyam district employment OGD resource
https://punjab.data.gov.in/resource/karnataka-udyam-registration-units-employment-details-district-wise-4

ASI Summary Results 2022-23
https://www.mospi.gov.in/sites/default/files/asi_results/ASI%20Summary%20Results%202022-23.pdf

NCVT MIS ITI Count
https://www.ncvtmis.gov.in/pages/iti/count.aspx

NCVT MIS district/trade extract evidence
https://ncvtmis.gov.in/Pages/FileDownload.aspx?fileId=1171

PLFS Annual Report 2022-23
https://www.mospi.gov.in/sites/default/files/publication_reports/AR_PLFS_2022_23N.pdf

Karnataka Udyam 2023-2025 catalog
https://kerala.data.gov.in/catalog/details-karnataka-udyam-registration-2023-2025

Safeguards: public, aggregated data only; no person-level records; human review before policy use; provenance retained; official-source IDs and checksums recorded; proxy-versus-actual status visible; no official KSI formula claimed.

AI-use disclosure: AI-assisted research and code generation were used in developing the prototype. All public-source claims used in the submission must be checked against the cited source documents before final filing.

Next upgrade: load the live LGD export, live NCVT MIS district/trade seats for the two pilot districts, and PLFS occupation-by-industry mix. Then rerun and lock the empirical results.